MAS Solution Approaches · Independent Review Record

Review method: Each handout is reviewed by an independent sub-agent. The reviewer did not take part in writing, works in a fresh context, and only reads without changing anything. Sources for verification include: the original "Specific behaviours" text in each question's marking key in `evidence/`, the full text of the 2016–2025 exam reports, the PDF pages of the original papers and keys, and independent verification using sympy/scipy/numpy.

Handling principles: The author reviews each review comment one by one. All high-severity comments are checked back against the original exam paper wording, and anything that can be verified by calculation has been recalculated. Items marked "doubtful" by the reviewer are changed only when evidence is found.

Outcome categories: Accepted = changed as suggested; Partly accepted = wording changed but the suggestion was not followed fully; Rejected = not changed, with reasons stated.

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01 Complex numbers (12 items: 12 accepted)

#	Severity	Issue	Resolution
1	High	Regular hexagon written incorrectly: in 2025F-Q7, z_0 is at the origin, the hexagon is not centred at O, and the product is $6r^5$, not r^5	Accepted: the author recalculated (moduli in order r, √3r, 2r, √3r, r) and corrected it; the question-setting skill was corrected at the same time
2	Medium-high	2025A-Q19(b) omits the periodic solutions n = 3,4,5 + 12k	Accepted
3	Medium	Maximum argument: sin θ = r/d does not give the argument	Accepted
4	Medium	2020A-Q10(b) ray endpoint i lies on the locus (filled point)	Accepted
5	Low	The shortest-distance formula holds only when the point lies outside the circle	Accepted
6	Low	"Always use polar form for multiplication and division" contradicts "use the conjugate for division"	Accepted
7	Low	Mismatch between report year and content (−z comes from the 2021 report)	Accepted
8	Low	The third-quadrant example is not the report's original wording; 5π/4 cannot be written either	Accepted
9	Low	Missing citation of 2022F-Q6; sample paper not specified	Accepted
10	Low	The "CF opener / CF hard question" labels do not match the cited CA questions	Accepted
11	Low	"i^n has period 4" is imprecise ($n \cdot i^n$ is not periodic)	Accepted
12	Low	When θ < 0, write clockwise by |θ|	Accepted

02 Functions and graphs (16 items: 16 accepted)

#	Severity	Issue	Resolution
1	High	In the reciprocal-function table, "f → ±∞ ⇒ horizontal asymptote y = 0" is wrong: a vertical asymptote of f corresponds to the open point (b, 0) of 1/f	Accepted: split into two rows
2	Medium	The table omits "horizontal asymptote y = c → y = 1/c"	Accepted
3	Medium	"Hole to hole" holds only when b ≠ 0	Accepted
4	Medium	2016A-Q12(b) is |f(x)| = d, not f(x) = d	Accepted

5	Medium	"Flat bottom of two V-shaped lines" is described incorrectly	Accepted: changed to the flat bottom of a sum of two absolute values; the author verified a = 3, k = 2
6	Low	The integral of the W-shaped graph is one triangle, not two	Accepted
7	Medium	The sign of the inverse function should be determined by ran f^{-1} = dom f	Accepted
8	Medium	When a square root is in the denominator, it must be > 0 (2025F-Q2(a))	Accepted
9	Medium	"c > 0 guarantees no zeros" is not enough; a and c must have the same sign	Accepted
10	Low	"Turning point (if the question requires it)": 2017F-Q5 awarded marks even though it was not required	Accepted
11	Low (doubtful)	The invented example "c = 2 touches the x-axis" conflicts with the meaning of c in the actual exam question	Accepted: replaced with the original wording from the exam question
12	Low	"Most stable across the whole course" is an overgeneralisation (in 2018 it was instead listed as well answered)	Accepted
13	Low	"Do not use it" omits the 2019 report	Accepted
14	Low	The example ln|x − 3| does not come from the 2022 report	Accepted
15	Low	Missing citations for 2016F-Q8(b), 2022F-Q2(a)	Accepted
16	Low	"Intersection point unchanged" is vague wording	Accepted

03 3D vectors and systems of linear equations (13 items: 13 accepted)

#	Severity	Issue	Resolution
1	High	Laser reflection direction written incorrectly: adding in the S→B direction gives a vector parallel to the mirror; it should be the sum of the unit vectors B→S and B→R, or đ_out − đ_in ∥ n	Accepted: the author used numpy to verify đ_out − đ_in = (0, 0.23, 0.46) ∥ (0, 1, 2)
2	High	2024A-Q16(c) is the shortest distance between two lines (two independent parameters); 2017A-Q14 is solving an inequality to find duration; the two were conflated	Accepted: split into two types
3	Medium	"CF is examined every year" is false (not examined in 2017 or 2019; in 2021 it was in CA)	Accepted
4	Medium	"Named five years in a row" does not match the reports; 2024 is missing	Accepted
5	Medium	"Must write in parametric form" is an overgeneralisation; the key accepts representation by a relationship between variables	Accepted
6	Medium	"Wavy lines, dot product, cross product" is not the original report wording	Accepted: marked as author's suggestion
7	Low-medium	The point-to-plane distance formula is not on the formula sheet	Accepted
8	Low	"Many people have never seen it" is an exaggerated paraphrase	Accepted
9	Low	The line-plane angle formula is missing an absolute value, and is inconsistent	Accepted
10	Low	The geometric interpretation table omits "three planes coincide"; 2021A-Q14 requires both points to be written	Accepted
11	Low	The letters a and d are each used for two different things	Accepted
12	Low	"CA final question" does not match 2018F-Q8	Accepted
13	Low (doubtful)	Sample paper question not specified	Accepted

04 Vector calculus and motion (12 items: 11 accepted, 1 partly accepted)

#	Severity	Issue	Resolution
1	High	2019A-Q13(d) is not a parabola; it is a figure-of-eight shape	Accepted
2	High	All three examples of restrictions on values do not match the key (2024A-Q18 requires $x \geq 2$ and $y \geq 0$)	Accepted: replaced with the original key wording
3	Medium	"One loop = 0 to 2π" is an overgeneralisation (2017A-Q15 is 6π, 2019A-Q13 is 4π)	Accepted
4	Medium	2021A-Q19 has no wind and no launch angle, but has lift	Accepted: split into two models
5	Medium	"Uniform velocity" should be "constant speed"	Accepted
6	Medium	The descent angle $90° - \varphi$ gives a negative value ($\varphi = 143.3°$)	Accepted: changed to $\lvert 90° - \varphi \rvert$; the author verified 53.3°
7	Low	"Three marking points" is an overgeneralisation	Accepted
8	Low	Height is relative to the centre; 100 m above the ground corresponds to $y = 20$	Accepted
9	Low	The order of subtraction has no source or rule given	Accepted
10	Low	In 2024A-Q16(c), both speed and departure time can be adjusted	Accepted
11	Low	Eliminating the parameter omits the type "x is linear in t" (named in the 2018 report)	Accepted
12	Low (doubtful)	SHM in 2019A-Q18(d) is not mentioned	Partly accepted: added a cross-reference; the method is in the differential equations handout

05 Integration techniques and applications (11 items: 11 accepted)

#	Severity	Issue	Resolution
1	Medium	"Irreducible quadratic" was paired with the reducible $x^2 + 3x + 1$ (discriminant $5 > 0$)	Accepted: replaced with $x^2 + 2$; a separate note was added for 2022F-Q4
2	Medium	The error example from the 2021 report is paraphrased incorrectly (the original is $\sqrt{(2 - \sin^2\theta)} \neq \sqrt{2} - \sin\theta$)	Accepted
3	Medium	"Substitution is CF every year" is false (not examined in 2022), and there are 7-mark questions	Accepted
4	Medium	"Partial fractions are examined every year" is false (not examined in 2017 or 2023)	Accepted
5	Medium	2023A-Q17(c) is an increment formula question; 2022A-Q17 does not use dV/dh	Accepted: split and rewritten
6	Medium	"The absolute value gets 1 mark on its own" is false; the absolute value is included in the antiderivative mark	Accepted
7	Low	The "correct notation" mark is awarded together with the limits; the area expression does not need π	Accepted
8	Low	2016A-Q9 is an indefinite integral; in 2021F-Q3 the substitution must be chosen	Accepted
9	Low	3 questions were not classified	Accepted
10	Low	The 4 in the cardioid comes from multiplying two parts	Accepted
11	Low	The 2024 report says the numerator after finding a common denominator was written incorrectly, not that comparing coefficients was wrong	Accepted

06 Rates of Change and Differential Equations (15 items: 15 adopted)

#	Severity	Issue	Action
1	Medium	The key's main solution for 2017A-Q17(b) is counting periods, not $\int\|v\|$	Adopted
2	High	The inflection point in 2025A-Q16 is at the maximum slope (3, −1), not where the slope is 0	Adopted
3	Medium	"10% growth per year" should be an instantaneous growth rate ($dP/dt = 0.1P$); otherwise the answer changes from 333 to about 412	Adopted: editor verified both results
4	Low	x(0) in 2025A-Q15 is −1	Adopted
5	Medium	The letter k refers to carrying capacity and the proportionality constant in different questions, and is easily confused	Adopted
6	Low	"Initial value above k/2 is concave down throughout" needs $P_0 < k$ added	Adopted
7	Low (uncertain)	SHM criterion should allow for a shifted equilibrium point $a = -n^2(x - c)$	Adopted
8	Low	"Cannot differentiate directly with respect to t" is unclear	Adopted
9	Low	2024A-Q12 did not give a direction; both ±A sin are awarded marks	Adopted
10	Medium	Marking-key reminder omitted 2024 (logistic), 2017 (standard equation), 2022 (absolute value)	Adopted
11	Low	The exact wording "integration statement" appears in only two years	Adopted
12	Low	2016A-Q18 is a − bx; the two conditions come from different sources	Adopted
13	Low	"Keep only one independent variable" contradicts the shadow question; the past paper list has incorrect inclusions and omissions	Adopted
14	Low	A vertical tangent only holds when the numerator ≠ 0	Adopted
15	Low	2018A-Q20 is "do not" solve, not "need not"	Adopted

07 Statistical Inference (16 items: 16 adopted)

#	Severity	Issue	Action
1	High	2022A-Q12(d) is "the same sample duplicated", not two samples combined, and conflicts with the conclusion of 2019A-Q15(c)	Adopted: the two cases written separately
2	Medium (uncertain)	"key accepts both" has no basis; key only gives $\bar{X} < T/n$	Adopted; question-setting skill corrected accordingly
3	Medium	"Examined every year" is untrue (not examined in 2016, 2020)	Adopted
4	Medium	Missed the counterexample 2025A-Q13(e) (many intervals not containing μ are evidence)	Adopted
5	Low	"Citing CLT" is not a separate scoring condition in the key	Adopted
6	Low	The calculation for 304.4 with 2.326 is inconsistent (should be 304.3)	Adopted: editor recalculated
7	Low	Rounding when back-solving for n: the key accepts both 2.576 and 2.58 differences; add 2024A-Q14(f)	Adopted
8	Low	Missed the one-tailed and half-interval cases	Adopted
9	Low	2024A-Q14(c) does not ask for n	Adopted
10	Low	"The answer does not change" only applies to 2018A-Q12(c)	Adopted
11	Low	The graphing question missed the mark for indicating the probability region	Adopted
12	Low	The key reason for 2024A-Q14(e) is different sample sizes	Adopted
13	Low	Sample paper question not explained, and sample paper A9 not cited	Adopted
14	Low	"Definitely" should be "very likely"	Adopted
15	Low	Precision table lacks context (μ = 80); s and σ used interchangeably	Adopted

16	Low	"n ≥ 30 applies" is an overgeneralisation; for a normal population any n works	Adopted

Summary (7 MAS documents): 95 comments in total (including 5 "uncertain"), 94 adopted, 1 partially adopted, 0 rejected. 8 high-severity items, all revised after returning to the original paper text or recalculating.

Classification corrections after user feedback (2026-10-09)

- User feedback: "Vectors" contained unrelated differential equation questions. Verified true: 2017A-Q18 (carousel, cosine rule + implicit differentiation) and 2019A-Q18 (Ferris wheel, related rates + SHM proof) contain no vector content; they were originally misclassified as vector calculus under "circular motion". Removed; kept only in "Rates of Change and Differential Equations"; the corresponding section in the notes was moved across as well. Vector calculus 14 → 12 questions.
- 2021A-Q19 (ski jump) gives the position vector r(t) in the question and asks for the Cartesian path and landing angle; this is vector motion and is retained.
- Keyword review of all tags; MAS found no other misclassifications.